

Finding Information About the File's Timestamps

Files have timestamps indicating when they were created, last modified, and last accessed. Python's `os.path.getctime()`, `os.path.getmtime()`, and `os.path.getatime()`, respectively, return these timestamps as floating-point values indicating the number of seconds since the *Unix epoch*, midnight on January 1, 1970, in the Coordinated Universal Time (UTC) time zone. Enter the following into the interactive shell:

Python

```
> import os
> filename = 'C:/Windows/system32/notepad.exe'
> os.path.getctime(filename)
1625705942.1165037
> os.path.getmtime(filename)
1625705942.1205275
> os.path.getatime(filename)
1631217101.8869188
```

These float values are easy for programs to use since they're just single numbers, but you'll need functions from Python's `time` module to make them simpler for humans to read. The `time.localtime()` function converts a Unix epoch timestamp into a `struct_time` object in the computer's time zone. A `struct_time` object has several attributes whose names begin with `tm_` for obtaining date and time information. Enter the following into the interactive shell:

Python

```
>>> import os
>>> filename = 'C:/Windows/system32/notepad.exe'
>>> ctimestamp = os.path.getctime(filename)
>>> import time
>>> time.localtime(ctimestamp)
time.struct_time(tm_year=2021, tm_mon=7, tm_mday=7, tm_hour=19,
tm_min=59, tm_sec=2, tm_wday=2, tm_yday=188, tm_isdst=1)
>>> st = time.localtime(ctimestamp)
>>> st.tm_year
2021
>>> st.tm_mon
7
```